

Report PCD 2025-2026

Clustered Smart Home Alarm System (Assignment 4 - Exercise 1)

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1 Introduction

The objective of this assignment is to extend the *Smart Home Alarm System* (SHAS) from Assignment 3 into a distributed system running on an **Apache Pekko Cluster**. The cluster architecture decouples physical deployment from application logic: sensors, keypads, sirens, and the control unit can reside on different cluster nodes, communicating transparently via message passing.

2 Analysis

Transitioning to a clustered environment introduces major concurrency and distribution challenges:

- **Location Transparency:** Actors (sensors, keypads, control units) cannot rely on local reference passing. They must discover each other dynamically across the cluster.
- **Partial Failures and Recovery:** If the Control Unit node crashes or restarts, its state is lost. From a security standpoint, the restarted node must not make assumptions about the system's previous state (neither armed nor disarmed), and it must enter a safe **RECOVERY** state where events are logged/ignored until a correct PIN is supplied.
- **Serialization:** Message objects are transmitted over network sockets, requiring a robust serialization mechanism instead of standard Java serialization.

3 Design

The system is built on Apache Pekko Typed Cluster using the following design strategies:

- **Dynamic Discovery via Receptionist:** Both the Control Unit and Siren register themselves with the Pekko `Receptionist` using dedicated `ServiceKey` instances. Sensors and Keypads subscribe to the Control Unit service key to dynamically track available references and broadcast commands.
- **Safe Recovery State:** The `ControlUnitActor` is structured as a typed state machine. On startup or restart, it enters the `RECOVERY` state. While in recovery, sensor activations are logged and discarded, and arming commands are rejected. A correct `PinSubmitted` command transitions the system back to `DISARMED`.
- **Jackson Serialization:** All cluster-transmissible messages implement the empty marker interface `MySerializable`. Pekko is configured via `application.conf` to bind this interface to the `jackson-json` serializer.
- **Optimized Cluster Join Configuration:** The seed nodes timeout (`seed-node-timeout`) is tuned to `1s` and `retry-unsuccessful-join-after` is set to `1s` to ensure fast cluster formation and reduce integration test delays.

4 Validation

Validation comprises unit tests, integration tests, and a self-contained automatic simulation:

1. **Unit Tests (`ControlUnitActorTest`):** Exercises the state machine in isolation, verifying that the initial state is `RECOVERY`, sensor events are ignored, and recovery completes only upon receiving the correct PIN.
2. **Integration Tests (`ClusteredSystemTest`):** Programmatically spawns three local `ActorSystem` nodes on different ports, forms a cluster, and simulates a full alarm arming, sensor activation (entry delay), and disarming workflow over the loopback network.
3. **Simulation Demo (`Main`):** Automatically initializes three nodes representing different physical components, demonstrates correct PIN disarming from recovery, arms the system, terminates the Control Unit node, spawns a new one to demonstrate recovery state activation, and verifies that sensors are ignored until the system is disarmed again.

5 Conclusions

The Pekko Cluster model enables building robust, secure, and location-transparent distributed applications. By pairing the Receptionist discovery mechanism with a secure recovery protocol, the system safely handles partial node failures and restarts without exposing the smart home to unauthorized access or false alarms.